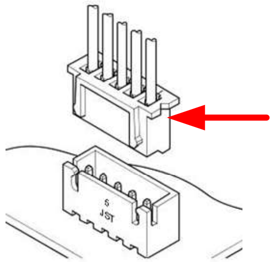
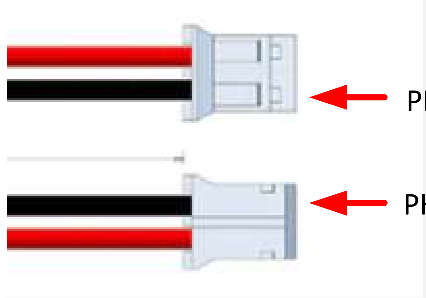
Buffer Connections

Items required for the buffer
1 x Mellow LLL Buffer
2 x JST PHR-2
1 x JST XH 3-Way
1 x JST XH 4-Way



XH Notch Pin 1

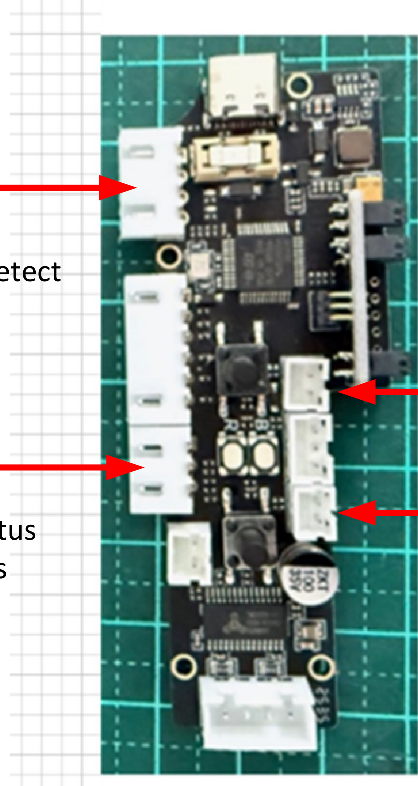


PH Pin 1

PH Pin 1

Note you will need cable and connector for your controll board

Control Board connections required
3 inputs,, Filament Detect , Button control Load/unload
2 outputs, control buffer load/unload



1)12/24
2)0V
3)Filament Detect
4)N/C

1)0V signal
2)Retract status
3)Feed Status

1) 0V
2) Filament Feed

1) 0V
2) Filament Retract

Example of connections to a Duet 6HC

From Buffer	**Operation**	**wire Col/Size**	**TO**
12/24V	Posative supply	Red/22	Posative 24V Supply
0V	0V supply	Black/22	Negative 24V DC supply
Filament Detect	Filament Detect Status	White/22	IO_8_in
Feed Status	Button signal Feed status	Blue/22	IO_6_in
Retract Status	Button singnal Retract status	Yellow/22	IO_7_in
0V signal	Gnd Signal	Black/22	IO7_GND
Filament Retract	Filament Retract	White /24	IO_7_out
Filament Feed	Filament Feed	Blue/24	IO_8_out

JST XH crimp wire size 22 > 26AWG
JST PH crimp wire size 24 > 30AWG
Stripping Length: 1.9 – 2.5 mm.

Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

Buffer Pin	Trigger Signal	Action Performed	Duration
Feed Control	Low Level	Buffer continuously feeds filament	Executes while the signal is held low
Retract Control	Low Level	Buffer continuously retracts filament	Executes while the signal is held low

Button	Operation	Signal Output (Buffer Pin)	Signal Type	Duration
Feed Button	Click	FEED pin outputs a high-level pulse	High Level	Automatically returns to low level after 3 seconds
Feed Button	Press & Hold	Continuous feeding	High Level	Until the button is released
Retract Button	Click	RETRACT pin outputs a low-level pulse	Low Level	Automatically returns to high level after 3 seconds
Retract Button	Press & Hold	Continuous retraction	Low Level	Until the button is released

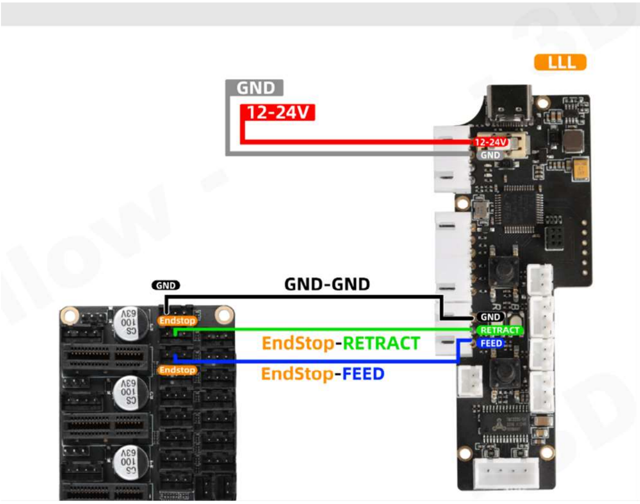
Note: The buffer stops the action when the signal returns to a high level.

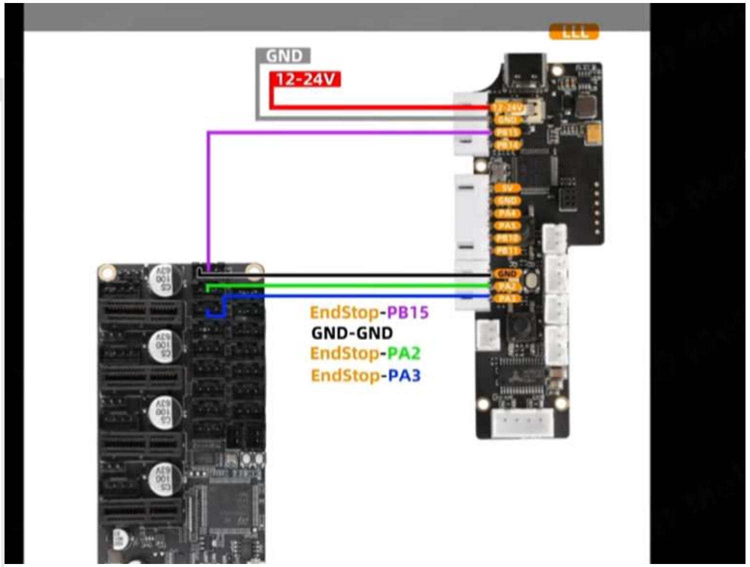
act

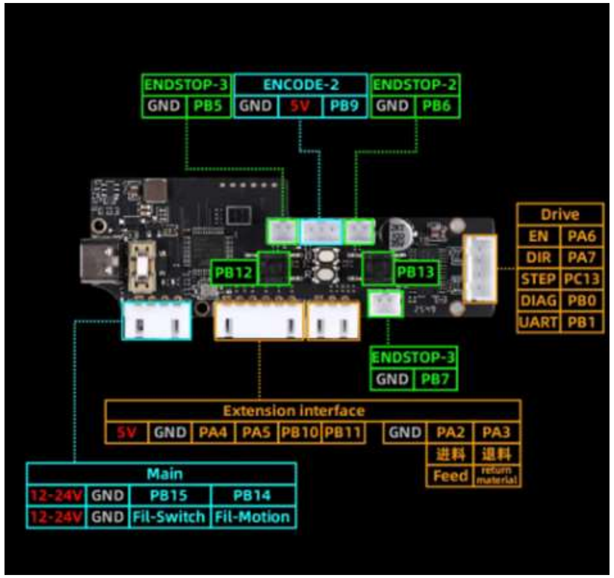
How to up date Firmware

https://mellow.klipper.cn/en/docs/ProductDoc/ExtensionBoard/fly-buffer-plus/update

https://drive.google.com/drive/folders/1wjAlmvND2FCBo72GI6J4LZEO3XSbTHVE







Signal Output Description

Button	Operation	Signal Output (Buffer Pin)	Signal Type	Duration
Feed Button (FEED)	Click	FEED pin outputs a high-level pulse	High Level	Automatically returns to low level after 3 seconds
Feed Button (FEED)	Long Press	Continuous feeding	High Level	Until the button is released
Retract Button (RETRACT)	Click	RETRACT pin outputs a low-level pulse	Low Level	Automatically returns to high level after 3 seconds
Retract Button (RETRACT)	Long Press	Continuous retraction	Low Level	Until the button is released

Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

Buffer Pin	Trigger Signal	Action Performed	Duration
PB5	Low Level	Buffer continuously feeds filament	Executes while the signal is held low
PB6	Low Level	Buffer continuously retracts filament	Executes while the signal is held low

Note: The buffer stops the action when the signal returns to a high level.